

Data Analytics and Business Intelligence English Participant Workbook

Practice pages for realistic field-specific meetings, pushback, documentation, and role-play preparation

Audience: data analysts, BI developers, analytics engineers, data scientists, reporting teams, business analysts, and data-driven managers

Focus: A data analytics and BI English curriculum for metric definitions, dashboards, data quality, stakeholder requests, causation, experimentation, governance, and insight presentation.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

How to Use This Workbook

For each module, read a workplace scenario and complete a bounded dialogue. Choose one of four options, then use the answer key to see why the selected phrase fits the evidence, risk, owner, and next step.

Module 1. Metric Definitions and Business Questions

Situation

Two teams use different definitions of active customer.

Stakeholder pressure: Pick one quickly for the dashboard.

Constraint: Business definition, source table, filter logic, and owner need agreement.

Terms to use

- metric
- business definition
- source of truth
- owner

Three-step decision practice

Situation

Two teams use different definitions of active customer.

Speaker	Line
Stakeholder	Pick one quickly for the dashboard.
ESL learner	Before we act, I need to confirm the _____ and how it affects business definition.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a metric definition note.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects business definition.

Option	Phrase
A	owner
B	evidence threshold
C	metric
D	source of truth

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Pick one quickly for the dashboard.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	approval path
B	operating constraint
C	decision owner
D	I understand the urgency. I need to verify metric and business definition against the stated constraint before I recommend the next step.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - Two teams use different definitions of active customer. - what should happen next?

Option	Phrase
A	approval path
B	operating constraint
C	decision owner
D	Review the stated constraint with the accountable owner, then prepare the metric definition note with the evidence, tradeoff, and requested decision.

Module 2. Data Quality and Trust

Situation

A dashboard shows a sudden drop after a pipeline change.

Stakeholder pressure: Tell users the data are wrong.

Constraint: Freshness, completeness, transformation logic, and upstream changes must be checked.

Terms to use

- data quality
- freshness
- completeness
- pipeline

Three-step decision practice

Situation

A dashboard shows a sudden drop after a pipeline change.

Speaker	Line
Stakeholder	Tell users the data are wrong.
ESL learner	Before we act, I need to confirm the _____ and how it affects freshness.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a data-quality incident update.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects freshness.

Option	Phrase
A	pipeline
B	data quality
C	freshness
D	completeness

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Tell users the data are wrong.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	evidence threshold
B	approval path
C	I understand the urgency. I need to verify data quality and freshness against the stated constraint before I recommend the next step.
D	I disagree, but I cannot name the boundary or the decision the group needs to make.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A dashboard shows a sudden drop after a pipeline change. - what should happen next?

Option	Phrase
A	Wait for the problem to become visible before assigning an owner or evidence source.
B	Review the stated constraint with the accountable owner, then prepare the data-quality incident update with the evidence, tradeoff, and requested decision.
C	Accept the request immediately because the stakeholder has already expressed urgency.
D	Send every available detail to senior leaders without identifying the decision they need to make.

Module 3. Dashboards and Executive Reporting

Situation

Executives ask for a dashboard with every possible metric.

Stakeholder pressure: Add all requested charts.

Constraint: Audience, decision, refresh cadence, and alert thresholds need focus.

Terms to use

- dashboard
- KPI
- threshold

- drill-down

Three-step decision practice

Situation

Executives ask for a dashboard with every possible metric.

Speaker	Line
Stakeholder	Add all requested charts.
ESL learner	Before we act, I need to confirm the _____ and how it affects KPI.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a dashboard requirements brief.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects KPI.

Option	Phrase
A	drill-down
B	evidence threshold
C	dashboard
D	threshold

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Add all requested charts.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	operating constraint
B	decision owner
C	We should act now and resolve dashboard after the result is visible.
D	I understand the urgency. I need to verify dashboard and KPI against the stated constraint before I recommend the next step.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - Executives ask for a dashboard with every possible metric. - what should happen next?

Option	Phrase
A	Accept the request immediately because the stakeholder has already expressed urgency.
B	Review the stated constraint with the accountable owner, then prepare the dashboard requirements brief with the evidence, tradeoff, and requested decision.
C	operating constraint

Option	Phrase
D	decision owner

Module 4. Causation, Correlation, and Caveats

Situation

Sales increased after a campaign and marketing claims causality.

Stakeholder pressure: Say the campaign caused the increase.

Constraint: Seasonality, selection bias, control groups, and other factors need analysis.

Terms to use

- correlation
- causation
- confounder
- selection bias

Three-step decision practice

Situation

Sales increased after a campaign and marketing claims causality.

Speaker	Line
Stakeholder	Say the campaign caused the increase.
ESL learner	Before we act, I need to confirm the _____ and how it affects causation.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a causality caveat paragraph.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects causation.

Option	Phrase
A	selection bias
B	evidence threshold
C	approval path
D	correlation

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Say the campaign caused the increase.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I understand the urgency. I need to verify correlation and causation against the stated constraint before I recommend the next step.
B	evidence threshold
C	approval path
D	operating constraint

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - Sales increased after a campaign and marketing claims causality. - what should happen next?

Option	Phrase
A	approval path
B	operating constraint
C	Review the stated constraint with the accountable owner, then prepare the causality caveat paragraph with the evidence, tradeoff, and requested decision.
D	evidence threshold

Module 5. SQL, Models, and Transformation Logic

Situation

A finance report differs from the warehouse model.

Stakeholder pressure: Fix the number silently.

Constraint: Join logic, grain, filters, and reconciliation need transparent review.

Terms to use

- SQL
- data model
- grain
- join logic

Three-step decision practice

Situation

A finance report differs from the warehouse model.

Speaker	Line
Stakeholder	Fix the number silently.
ESL learner	Before we act, I need to confirm the _____ and how it affects data model.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a reconciliation explanation.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects data model.

Option	Phrase
A	SQL
B	evidence threshold
C	approval path
D	operating constraint

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Fix the number silently.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I disagree, but I cannot name the boundary or the decision the group needs to make.
B	evidence threshold
C	approval path
D	I understand the urgency. I need to verify SQL and data model against the stated constraint before I recommend the next step.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A finance report differs from the warehouse model. - what should happen next?

Option	Phrase
A	Send every available detail to senior leaders without identifying the decision they need to make.
B	Wait for the problem to become visible before assigning an owner or evidence source.
C	evidence threshold
D	Review the stated constraint with the accountable owner, then prepare the reconciliation explanation with the evidence, tradeoff, and requested decision.

Module 6. Experiment Readouts and Statistical Thinking

Situation

A test shows a small lift with uncertain confidence.

Stakeholder pressure: Recommend full rollout.

Constraint: Sample size, effect size, confidence, guardrails, and cost should guide decisions.

Terms to use

- sample size
- effect size
- confidence interval

- guardrail

Three-step decision practice

Situation

A test shows a small lift with uncertain confidence.

Speaker	Line
Stakeholder	Recommend full rollout.
ESL learner	Before we act, I need to confirm the _____ and how it affects effect size.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a experiment readout slide.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects effect size.

Option	Phrase
A	evidence threshold
B	approval path
C	operating constraint
D	sample size

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Recommend full rollout.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	A experiment readout slide is unnecessary; verbal assurance should be enough.
B	I understand the urgency. I need to verify sample size and effect size against the stated constraint before I recommend the next step.
C	decision owner
D	We should act now and resolve sample size after the result is visible.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A test shows a small lift with uncertain confidence. - what should happen next?

Option	Phrase
A	Send every available detail to senior leaders without identifying the decision they need to make.
B	Wait for the problem to become visible before assigning an owner or evidence source.
C	evidence threshold

Option	Phrase
D	Review the stated constraint with the accountable owner, then prepare the experiment readout slide with the evidence, tradeoff, and requested decision.

Module 7. Data Governance and Privacy

Situation

A team requests customer-level data for broad exploration.

Stakeholder pressure: Send the extract.

Constraint: Purpose limitation, access control, retention, and privacy rules need confirmation.

Terms to use

- data governance
- privacy
- access control
- retention

Three-step decision practice

Situation

A team requests customer-level data for broad exploration.

Speaker	Line
Stakeholder	Send the extract.
ESL learner	Before we act, I need to confirm the _____ and how it affects privacy.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a data access response.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects privacy.

Option	Phrase
A	retention
B	evidence threshold
C	approval path
D	data governance

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Send the extract.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	We should act now and resolve data governance after the result is visible.
B	A data access response is unnecessary; verbal assurance should be enough.
C	I understand the urgency. I need to verify data governance and privacy against the stated constraint before I recommend the next step.
D	decision owner

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A team requests customer-level data for broad exploration. - what should happen next?

Option	Phrase
A	decision owner
B	Accept the request immediately because the stakeholder has already expressed urgency.
C	Review the stated constraint with the accountable owner, then prepare the data access response with the evidence, tradeoff, and requested decision.
D	operating constraint

Module 8. Insight Storytelling and Recommendations

Situation

A stakeholder wants a simple recommendation from messy data.

Stakeholder pressure: Hide limitations to make the story clear.

Constraint: Uncertainty, tradeoffs, assumptions, and next tests need visible framing.

Terms to use

- insight
- recommendation
- assumption
- next best action

Three-step decision practice

Situation

A stakeholder wants a simple recommendation from messy data.

Speaker	Line
Stakeholder	Hide limitations to make the story clear.
ESL learner	Before we act, I need to confirm the _____ and how it affects recommendation.
Stakeholder	What would let us move forward responsibly?
ESL learner	I will test the constraint with the decision owner and return with a insight recommendation memo.

1. Complete the field language

Choose the term that belongs in this decision point.

Before we act, I need to confirm the _____ and how it affects recommendation.

Option	Phrase
A	next best action
B	insight
C	recommendation
D	assumption

2. Choose the strongest professional response

Select the response that preserves the relationship while setting an evidence-based boundary.

A stakeholder says, 'Hide limitations to make the story clear.' Which response gives a useful, professional form of pushback?

Option	Phrase
A	I disagree, but I cannot name the boundary or the decision the group needs to make.
B	I understand the urgency. I need to verify insight and recommendation against the stated constraint before I recommend the next step.
C	We should act now and resolve insight after the result is visible.
D	A insight recommendation memo is unnecessary; verbal assurance should be enough.

3. Select the next decision move

Choose the action that turns the discussion into a controlled workplace decision.

Given this situation - A stakeholder wants a simple recommendation from messy data. - what should happen next?

Option	Phrase
A	Review the stated constraint with the accountable owner, then prepare the insight recommendation memo with the evidence, tradeoff, and requested decision.
B	Send every available detail to senior leaders without identifying the decision they need to make.
C	Wait for the problem to become visible before assigning an owner or evidence source.
D	evidence threshold

Answer Key and Rationale

Check each stage after completing the practice sequence. The rationale explains the decision skill the strongest response demonstrates.

Metric Definitions and Business Questions - 1. Complete the field language

Correct answer	Why it fits
C. metric	'metric' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Metric Definitions and Business Questions - 2. Choose the strongest professional response

Correct answer	Why it fits
D. I understand the urgency. I need to verify metric and business definition against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Metric Definitions and Business Questions - 3. Select the next decision move

Correct answer	Why it fits
D. Review the stated constraint with the accountable owner, then prepare the metric definition note with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Data Quality and Trust - 1. Complete the field language

Correct answer	Why it fits
B. data quality	'data quality' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Data Quality and Trust - 2. Choose the strongest professional response

Correct answer	Why it fits
C. I understand the urgency. I need to verify data quality and freshness against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Data Quality and Trust - 3. Select the next decision move

Correct answer	Why it fits
B. Review the stated constraint with the accountable owner, then prepare the data-quality incident update with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Dashboards and Executive Reporting - 1. Complete the field language

Correct answer	Why it fits
C. dashboard	'dashboard' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Dashboards and Executive Reporting - 2. Choose the strongest professional response

Correct answer	Why it fits
D. I understand the urgency. I need to verify dashboard and KPI against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Dashboards and Executive Reporting - 3. Select the next decision move

Correct answer	Why it fits
B. Review the stated constraint with the accountable owner, then prepare the dashboard requirements brief with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Causation, Correlation, and Caveats - 1. Complete the field language

Correct answer	Why it fits
D. correlation	'correlation' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Causation, Correlation, and Caveats - 2. Choose the strongest professional response

Correct answer	Why it fits
A. I understand the urgency. I need to verify correlation and causation against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Causation, Correlation, and Caveats - 3. Select the next decision move

Correct answer	Why it fits
C. Review the stated constraint with the accountable owner, then prepare the causality caveat paragraph with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

SQL, Models, and Transformation Logic - 1. Complete the field language

Correct answer	Why it fits
A. SQL	'SQL' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

SQL, Models, and Transformation Logic - 2. Choose the strongest professional response

Correct answer	Why it fits
D. I understand the urgency. I need to verify SQL and data model against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

SQL, Models, and Transformation Logic - 3. Select the next decision move

Correct answer	Why it fits
D. Review the stated constraint with the accountable owner, then prepare the reconciliation explanation with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Experiment Readouts and Statistical Thinking - 1. Complete the field language

Correct answer	Why it fits
D. sample size	'sample size' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Experiment Readouts and Statistical Thinking - 2. Choose the strongest professional response

Correct answer	Why it fits
B. I understand the urgency. I need to verify sample size and effect size against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Experiment Readouts and Statistical Thinking - 3. Select the next decision move

Correct answer	Why it fits
D. Review the stated constraint with the accountable owner, then prepare the experiment readout slide with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Data Governance and Privacy - 1. Complete the field language

Correct answer	Why it fits
D. data governance	'data governance' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Data Governance and Privacy - 2. Choose the strongest professional response

Correct answer	Why it fits
C. I understand the urgency. I need to verify data governance and privacy against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Data Governance and Privacy - 3. Select the next decision move

Correct answer	Why it fits
C. Review the stated constraint with the accountable owner, then prepare the data access response with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.

Insight Storytelling and Recommendations - 1. Complete the field language

Correct answer	Why it fits
B. insight	'insight' is the field-specific variable that must be checked before the team can proceed responsibly. This is the decision variable the learner must confirm before the team acts.

Insight Storytelling and Recommendations - 2. Choose the strongest professional response

Correct answer	Why it fits
B. I understand the urgency. I need to verify insight and recommendation against the stated constraint before I recommend the next step.	The strongest response names the field terms, the constraint, and the condition for a recommendation without sounding defensive or passive. This response is specific, respectful, and appropriately conditional. It converts pressure into a controlled decision.

Insight Storytelling and Recommendations - 3. Select the next decision move

Correct answer	Why it fits
A. Review the stated constraint with the accountable owner, then prepare the insight recommendation memo with the evidence, tradeoff, and requested decision.	The best next move makes the evidence, tradeoff, owner, and requested decision visible in a concise workplace artifact. This sequence creates a concise, decision-ready artifact with a clear owner and evidence boundary.